

A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization

Julius A. Zeiss¹

joint work with Mario Berta¹, Omar Fawzi², Gereon Koßmann¹ and Martin Plávala³

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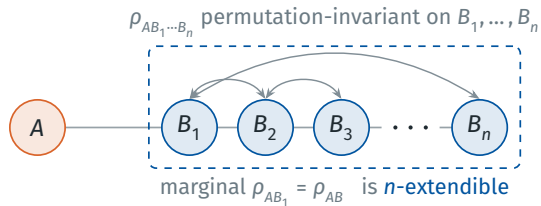
² Univ Lyon, Inria, ENS Lyon, UCBL, LIP, Lyon, France ³ Leibniz Universität Hannover, Germany

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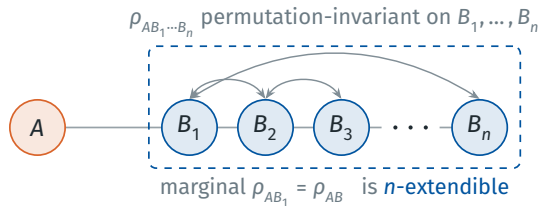
[ZKFB '25] [arXiv:2507.12302](https://arxiv.org/abs/2507.12302) · [KZFB '25] [arXiv:2507.12326](https://arxiv.org/abs/2507.12326)

[ZKSP '26] [arXiv:2601.15184](https://arxiv.org/abs/2601.15184) (with R. Schwonnek) · [KZ '26] [arXiv:2607.23689](https://arxiv.org/abs/2607.23689)

Quantum de Finetti theorems: symmetry $\Rightarrow$ independence



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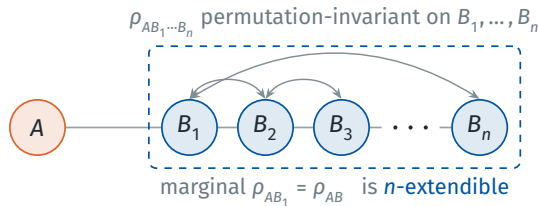
Asymptotic. extendible for all $n \Rightarrow$

$\rho_{AB} \in \text{SEP}$ exactly:

$\rho_{AB} = \sum_x p(x) \rho_A^x \otimes \rho_B^x$, a mixture of products.

[Hudson–Moody '76, Raggio–Werner '89]

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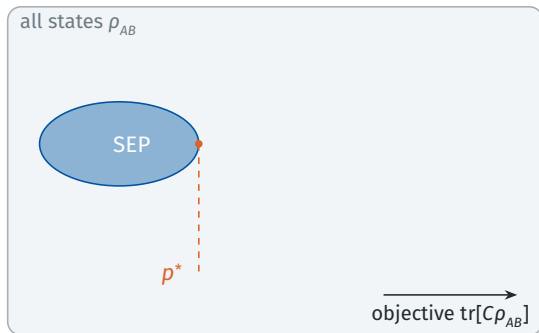
FOCUS OF THIS TALK

Finite. n -extendible $\Rightarrow \exists \sigma_{AB} \in \text{SEP}$:

$\|\rho_{AB} - \sigma_{AB}\|_1 \leq \varepsilon(n, |A|, |B|)$ with $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$.

Monogamy. B cannot be strongly correlated with A and exchangeable with $n - 1$ further copies: symmetry forces independence up to classical correlations.

Why finite: separability problems and their relaxations



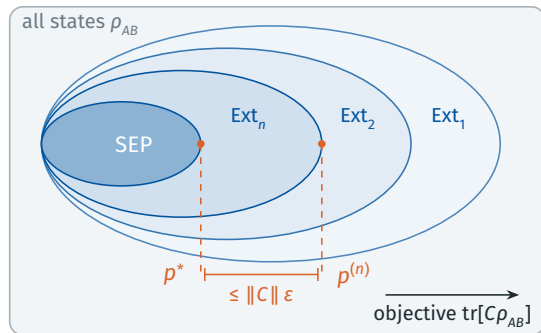
Separability problem:

$$p^* = \max_{\rho_{AB} \in \text{SEP}} \text{tr}[C\rho_{AB}]$$

linear objective, hard feasible set

[Gurvits '03]

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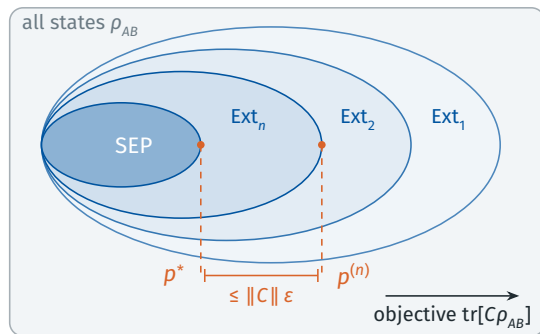
[Gurvits '03]

Relaxation: $p^{(n)} = \max_{\rho_{AB} \in \text{Ext}_n} \text{tr}[C\rho_{AB}]$,

an SDP with $p^{(1)} \geq p^{(2)} \geq \dots \geq p^*$

[Doherty–Parrilo–Spedalieri '02]

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[Doherty–Parrilo–Spedalieri '02]

Finite de Finetti $\implies 0 \leq p^{(n)} - p^* \leq \|C\|_{\infty} \epsilon(n, |A|, |B|)$:

level $n(\delta)$ reaches accuracy δ – the asymptotic theorem gives no $n(\delta)$.

Adding constraints: constrained separability

$\text{cSEP} = \text{conv}\{\rho_A \otimes \rho_B : \rho_A \in C_A, \rho_B \in C_B\}$ – constraints act on the *factors* [BBFS '21; ZKFB '25]

$C_B = \{\rho_B : \Phi_B(\rho_B) = W_B \otimes \rho_{B'}\}$ – a linear map of ρ_B equals a fixed operator $\otimes$ a marginal of ρ_B ; likewise C_A

covers fixed-point constraints (e.g. symmetries) and fixed-marginal constraints

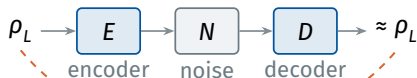
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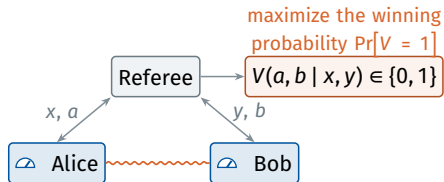
Error correction [BBFS '21; KZFB '25]



maximize the fidelity

bilinear in (E, D) · Choi constraints

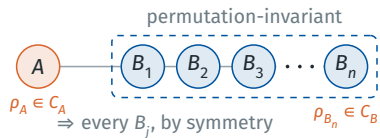
Non-local games [ZKFB '25]



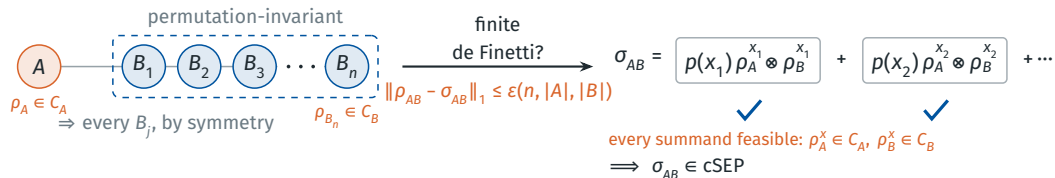
shared entangled state
of fixed dimension

bilinear in (state, measurements) · no-signalling

What we are looking for

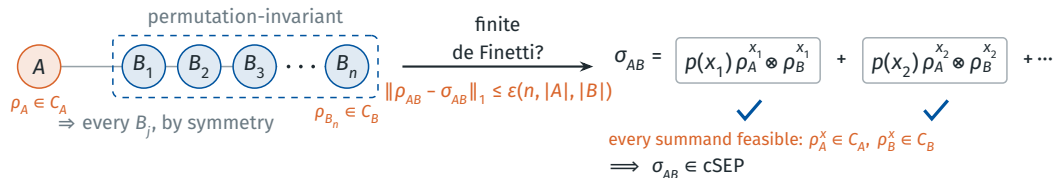


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Wanted: a finite de Finetti theorem with $\sigma_{AB} \in \text{cSEP}$ – every product term feasible, not only their mixture.

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The symmetric-subspace proofs with the **best rate**, $O(|B|^2/n)$

[Christandl–König–Mitchison–Renner '07; Harrow '13], cannot deliver this: there the constraints hold **only on average**.

The resolution: an information-theoretic de Finetti theorem

1 Measure

IC measurement

$M_{B \rightarrow Z}$ on every B_j

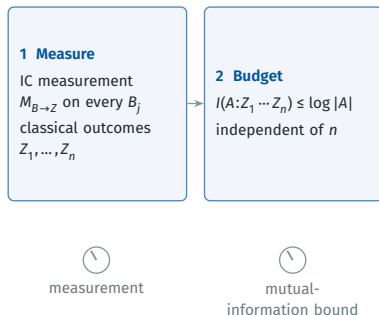
classical outcomes

$Z_1, \dots, Z_n$

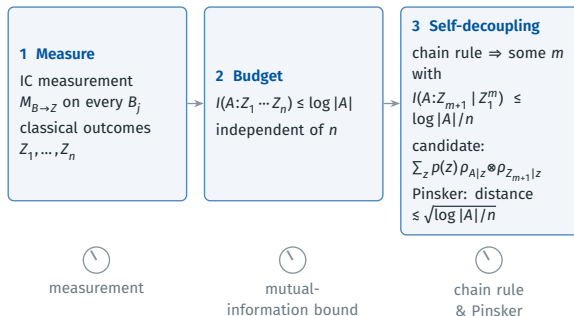


measurement

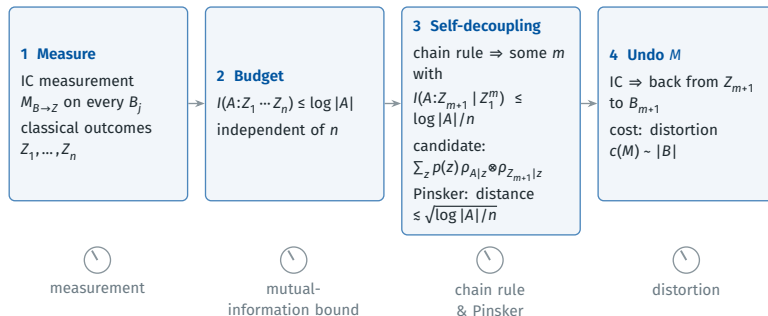
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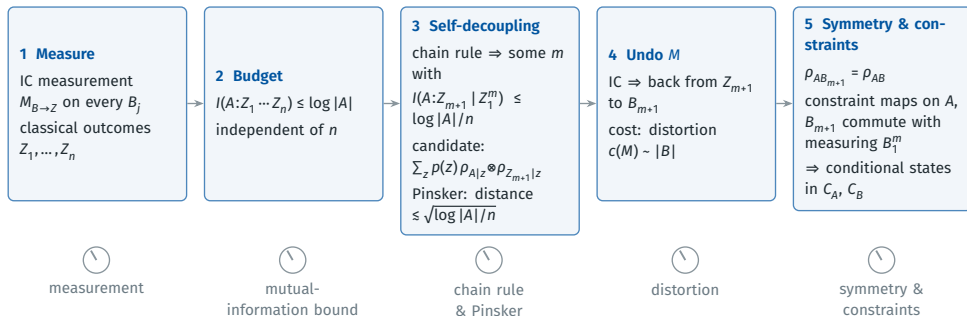
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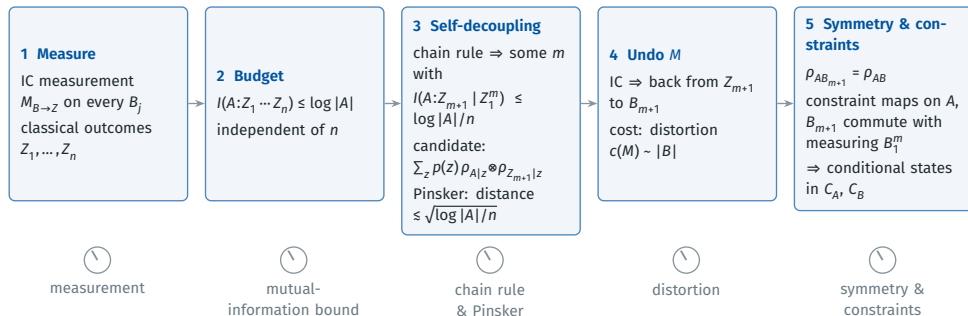
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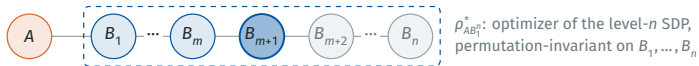
The resolution: an information-theoretic de Finetti theorem



Result: $\|\rho_{AB} - \sigma_{AB}\|_1 \leq c(M) \sqrt{2 \ln 2 \log |A| / n}$ with $\sigma_{AB} = \sum_{z_1^m} p(z_1^m) \rho_{A|z_1^m} \otimes \rho_{B_{m+1}|z_1^m} \in \text{cSEP}$
[Berta–Borderi–Fawzi–Scholz '21, Thm. 2.3]

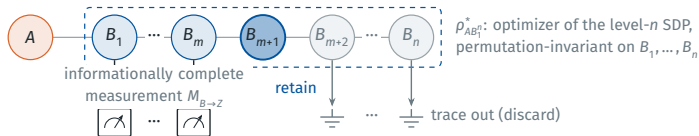
The proof is an algorithm: measurement-based rounding

Run the proof on the optimizer: measure, condition, average – it *returns* a point of cSEP.



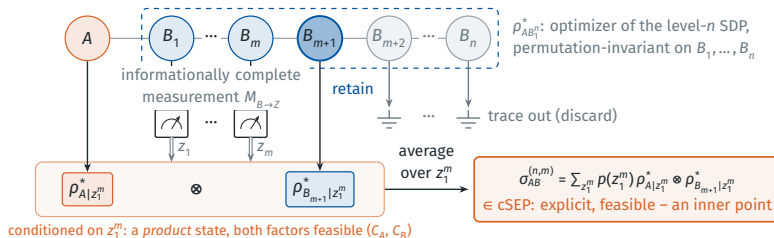
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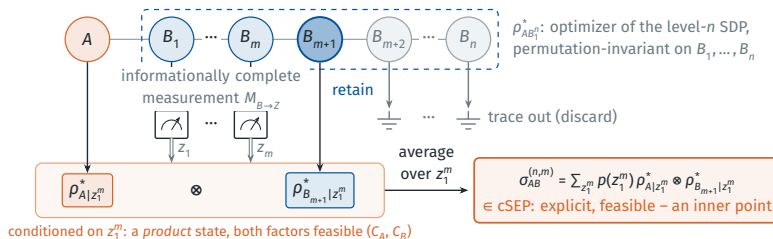
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Guarantee [ZKFB '25; KZFB '25] the rounded state is a **certifiably good explicit strategy** – it sandwiches the optimum with the SDP value:

$$\underbrace{\text{tr}[C \sigma_{AB}^{(n)}]}_{\text{inner: explicit strategy}} \leq p^* \leq \underbrace{p^{(n)}}_{\text{outer: SDP}} \quad \text{with gap} \leq \|C\| \varepsilon(n, |A|, |B|) \rightarrow 0$$

Schur–Weyl duality makes the rounding **efficient**: $\text{poly}(1/\varepsilon)$ arithmetic operations [KZ '26].

Every proof step is a dial

Three questions after the warm-up:

- Can we **exploit** the computationally costly extension symmetry?
- What if the problem has **more structure** – fixed points $\Phi(\rho) = \rho$?
- What if we have **more general structures** – convex bodies in GPTs?

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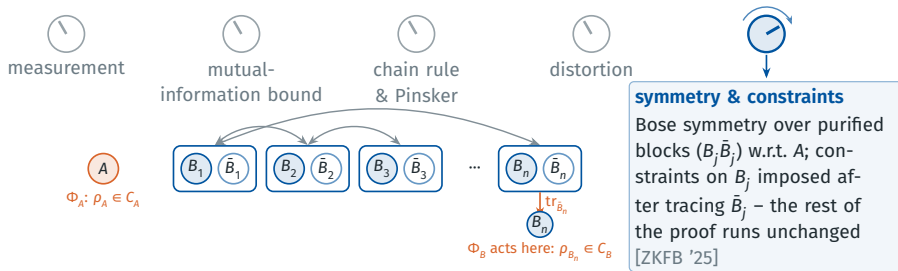
	 measurement	 mutual- information bound	 chain rule & Pinsker	 distortion	 symmetry & constraints	
Bose-symmetric de Finetti	☐	☐	☐	☐	☒	[ZKFB '25]
fixed-point de Finetti	☒	☒	☒	☒	☒	[KZ '26]
beyond quantum theory	☒	☒	☒	☒	☒	[ZKSP '26]

Turning the dials: Bose-symmetric de Finetti

Q. Can we exploit the computationally costly extension symmetry?

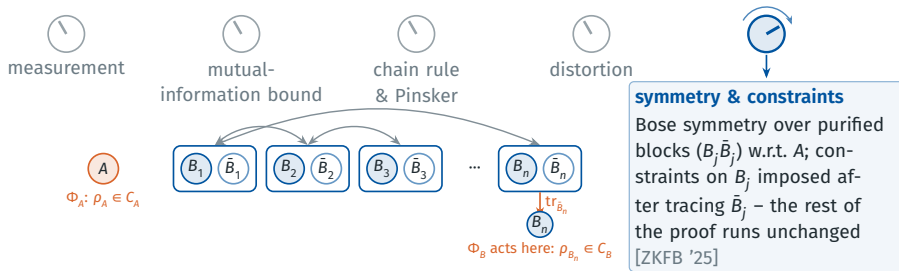
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Theorem [ZKFB '25] the bound survives with $\sigma_{AB} \in \text{cSEP}$: $\|\rho_{AB} - \sigma_{AB}\|_1 \leq c(M)\sqrt{2 \ln 2 \log |A|/n}$.

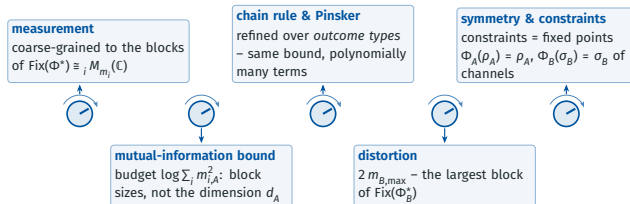
- (i) **symmetric-subspace methods extend to constraints** – term-wise, not only on average.
- (ii) an **exponential speed-up** of the outer approximation algorithm.

Turning the dials: fixed-point de Finetti

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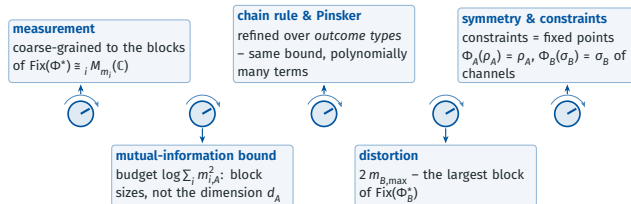
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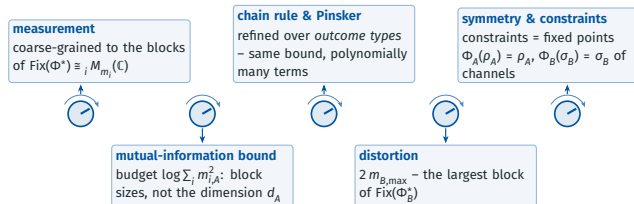


Theorem [KZ '26] if the n -extension is fixed by Φ_A on A and by Φ_B on each B_j (full-rank fixed points), then some $\tilde{\rho}_{AB} \in \text{conv}\{\rho_A \otimes \sigma_B : \text{both factors fixed}\}$ – term-wise ✓ – satisfies

$$\|\rho_{AB} - \tilde{\rho}_{AB}\|_1 \leq \underbrace{2 m_{B,\max}}_{\text{distortion}} \sqrt{\underbrace{2 \ln 2 \log(\sum_i m_{i,A}^2)}_{\text{budget}} / n}$$

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next: reading this as an **interpolation** – then pushing the machinery **beyond**: double-sided extensions

Fixed-point de Finetti: an interpolation classical $\leftrightarrow$ quantum

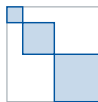
The structure. the observables fixed by the channel form a **block algebra**, $\text{Fix}(\Phi^*) \cong \bigoplus_i M_{m_i}(\mathbb{C})$ – its block sizes m_i measure **how quantum** the constrained system still is; they alone enter the bound:

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classical: dephasing
 d blocks of size 1
 $m_{\max} = 1$ – no d -factor



in between: e.g. a twirl
blocks = irreps
 S_n -twirl: $\log n$ budget



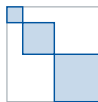
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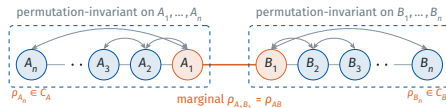
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the dimension dependence is not about d – it is about how quantum the constraints leave the system;

and the S_n -twirl entry is exactly what we exploit next

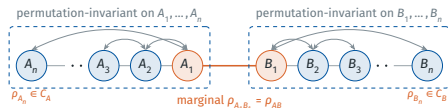
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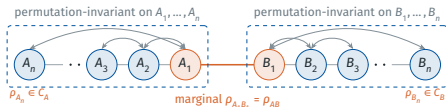


budget – the fixed-point machinery: the S_n -twirl on the A -chain $\Rightarrow (d_A^2 - 1) \log(n + d_A) - \log n$, not n

rate – the new ingredient: double chain rule, both chains measured $\Rightarrow n^2$ summands $\Rightarrow \sqrt{\log n} / n$

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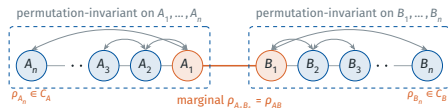
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one-sided [BBFS '21]

$1/\sqrt{n}$

term-wise ✓

one-and-a-half [CKMR '07]

$1/n$

on average only

double-sided [KZ '26]

$\sqrt{\log n} / n$

term-wise ✓ – **gap closed**

polynomial order matches [CKMR '07]: $n^{-1+o(1)}$; the $\sqrt{\log n}$ is open

Turning the dials: beyond quantum theory

 **gbit**
PR box

 **pentagon**
self-dual

 **convex body**
any K_A, K_B

Q. What if we have more general structures – convex bodies in GPTs?

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measurement

affine maps into a simplex,
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mutual-information bound

cone relative entropy
[Frenkel '23]: $I(A:B_1^n) \leq c(A)$ –
monogamy in every GPT



chain rule & Pinsker

classical after measuring; a
Pinsker inequality for cones



distortion

new measure-and-prepare
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symmetry & constraints

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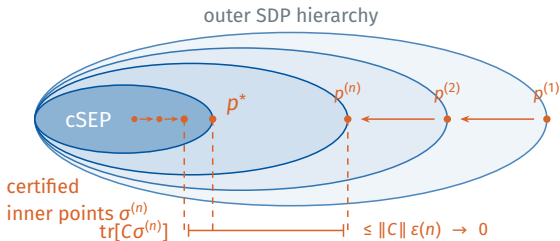


Theorem [ZKSP '26] every symmetric n -extendible $x_{AB} \in K_A \otimes_{\max} K_B$ is $2c(A, B)/\sqrt{n}$ -close to $K_A \otimes_{\min} K_B$ – for any convex bodies K_A, K_B .

$\Rightarrow$ convergent hierarchy with inequality constraints · certified inner points · LPs / conic programs

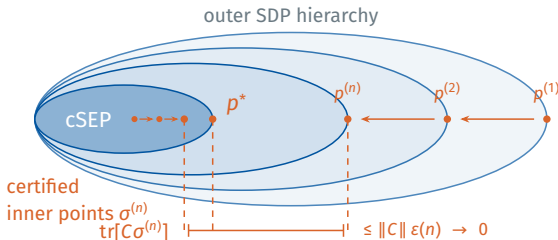
Summary

Closing the gap between SEP and cSEP: outer SDPs *and* certified inner points sandwich p^* in $\text{poly}(1/\varepsilon)$ time – performance certified by a [family of finite de Finetti theorems](#) [ZKFB '25; KZ '26]



Summary

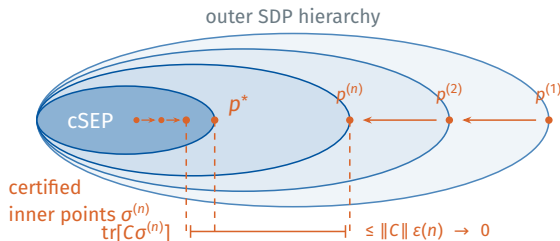
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Open: optimal cSEP rates (SEP: argmax rounding [JWX '26]) · coupled constraints beyond quantum

Thank you.

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[ZKSP '26] [arXiv:2601.15184](#) · [KZ '26] [arXiv:2607.23689](#)

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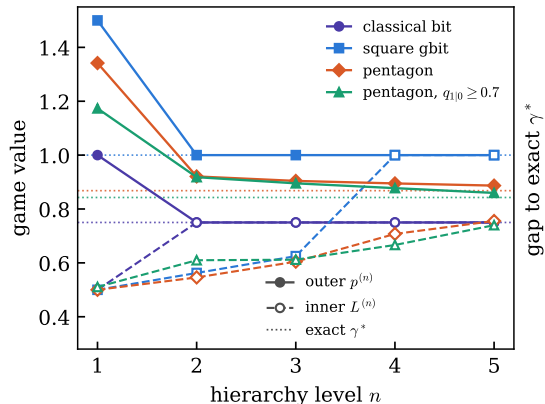
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Appendix: numerics – CHSH over polygon state spaces



CHSH over K_B = classical bit, square gbit, self-dual pentagon (also with a local inequality constraint $q_{1|0} \geq 0.7$); polytopes $\Rightarrow$ every level is an exact LP [ZKSP '26].

outer levels $p^{(n)}$ (filled) $\searrow$ exact γ^* (dotted)
 $\nwarrow$ rounded inner values $L^{(n)}$ (open)

bracket closes at $n = 2$ (bit, $\gamma^* = 3/4$) and $n = 4$ (gbit, $\gamma^* = 1$: the PR box); pentagon:
 $\gamma^* = (2\sqrt{5} - 1)/4$, gap $\approx 0.1/n$.